

Fujisaki Okamoto on el gamal

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Objective

Transform a CPA secure encryption scheme $P.\Sigma$ into a CCA secure encryption scheme $FO.\Sigma$

1 Overview

The Fujisaki-Okamoto (FO) transform converts a CPA-secure public-key encryption scheme into a CCA-secure one by adding:

- Randomness extraction using hash functions
- Integrity verification to detect tampering
- Re-encryption check to ensure consistency

2 Example: Transforming El-gamal

Alice(Sender)

Bob(receiver)

- Choose $x \in \mathbb{Z}_q$
- Choose $h = g^x$.

$\leftarrow \leftarrow \text{Publickey:h} \leftarrow$

- Message m
- Choose $r \in \mathbb{Z}_q$
- $c_1 = g^r$
- $c_2 = m.h^r$

$\longrightarrow \xrightarrow{c_1, c_2} \longrightarrow$

- Receive (c_1, c_2)
- $m = \frac{c_2}{c_1^x}$
- Output: m

2.1 Problems with Malleability

ElGamal is multiplicatively malleable. Given a ciphertext $(c_1, c_2) = (g^r, m \cdot h^r)$ encrypting message m , an attacker can:

- Multiply by a constant: Create $(c_1, k \cdot c_2)$ which decrypts to $k \cdot m$
- Combine ciphertexts: Given two ciphertexts encrypting m_1 and m_2 , create their "product" encrypting $m_1 \cdot m_2$
- Re-randomize: Add fresh randomness to create a different ciphertext for the same message

2.2 Concrete Attack Example

- Original ciphertext: $(g^r, m \cdot h^r)$ encrypting message m
- Attacker creates: $(g^r, 2 \cdot m \cdot h^r)$ by multiplying c_2 by 2
- This new ciphertext decrypts to $2m$
- In a CCA setting, the attacker could query this modified ciphertext to a decryption oracle and learn $2m$, then deduce m

This violates CCA security because the attacker can create valid ciphertexts related to the challenge ciphertext and use the decryption oracle to extract information.

2.3 Fujisaki Okamoto transform of El-gamal by Claude, (seems wrong)

Alice(Sender)

Bob(receiver)

- Choose $x \in \mathbb{Z}_q$
- Compute $h = g^x$

$\xleftrightarrow{\text{Publickey: } h}$
 $\xleftrightarrow{H_1, H_2}$

• ENCRYPTION

- Message: m
- Choose $\sigma \in \{0, 1\}^n$
- $r = H_1(m || \sigma)$
- $c_1 = g^r$
- $c_2 = (m \oplus H_2(\sigma || c_1) || \sigma) h^r$

$\xrightarrow{(c_1, c_2)}$

• DECRYPTION

- Receive (c_1, c_2)
- $(m' || \sigma') = \frac{c_2}{(c_1)^x}$
- Verify: $c_1 \stackrel{?}{=} g^{H_1(m' || \sigma')}$
- If NO: output **null**
- If YES: output $m' \oplus H_2(\sigma' || c_1)$
- READ THE SECOND POINT BELOW

2.3.1 Observations

- During decryption, $m' = m \oplus H_1(\sigma || c_1)$, and $\sigma' = \sigma$, so then $c_1 \neq g^{H_1(m' || \sigma')}$, unless the Hash function is NOT second pre image resistant. Absurd!
- One thing that could explain this. After $\frac{c_2}{(c_1)^x}$, we get $m' || \sigma'$, where $\sigma' = \sigma$ and $m' = m \oplus H_2(\sigma' || c_1)$ and we already have c_1 , so we can get $H_2(\sigma || c_1)$, hence we can decrypt c_2 here itself, and get m . Then for verification of this step: $c_1 \stackrel{?}{=} g^{H_1(m || \sigma)}$ // , that is what

is wrong above, using m' instead of m , but if we use m , then the verification succeeds.

- Receive (c_1, c_2)
 - $(m' || \sigma) = \frac{c_2}{(c_1)^x}$
 - $\mathbf{m} = \mathbf{m}' \oplus H_2(\sigma || c_1)$, the same σ we got from previous step.
 - Verify: $c_1 \stackrel{?}{=} g^{H_1(m || \sigma)}$
 - If NO: output **null**
 - If YES: output **m**
- One more issue is the usage of **message m** inside H_1 , which is not done in my version.

2.4 FJOK transform of El-gamal, my appraoch

Alice(Sender)

Bob(receiver)

- Choose $\alpha \in \mathbb{Z}_p$
- Compute $h = g_\alpha$

Public key: h

H_1, H_2

- Message: m
- Choose $x \in \{0, 1\}^n$
- $c_1 = (g^r, [x || H_1(x)] \oplus h^r)$
- $k = H_2(x)$
- $c_2 = m \oplus k$

(c_1, c_2)

- **DECRYPTION**
- Receive (c_1, c_2) , let $(v_1, v_2) = c_1$
- $[x || H_1(x)] = \frac{v_2}{(v_1)^\alpha}$, extract x , call it x' .
- IF $H_1(x') \neq H_1(x)$, then w.h.p. $x \neq x'$, proceed
- $k = H_2(x)$
- $m = c_2 \oplus k$

3 Why This Achieves CCA Security (Everything from here is for CLAUDE version)

3.1 Detailed Malleability Problem in ElGamal

3.1.1 Why Standard ElGamal Fails Against CCA Attacks

The Multiplicative Structure Problem:

Standard ElGamal has the property that if (c_1, c_2) encrypts message m , then:

$(c_1, k \cdot c_2)$ encrypts message $k \cdot m$ for any constant k $(c_1 \cdot c_1', c_2 \cdot c_2')$ encrypts $m \cdot m'$ if (c_1, c_2) encrypts m and (c_1', c_2') encrypts m' .

3.2 Concrete CCA Attack on Standard ElGamal:

Setup: Attacker wants to break message m^* encrypted as $(c_1^*, c_2^*) = (g^{r^*}, m^* \cdot h^{r^*})$ Attack Steps:

- Choose a known value $k \neq 1$ (e.g., $k = 2$)
- Create modified ciphertext: $(c_1^*, k \cdot c_2^*) = (g^{r^*}, k \cdot m^* \cdot h^{r^*})$
- Submit this to the decryption oracle
- Oracle returns $k \cdot m^*$ (since it's a valid ciphertext, just not the original)
- Compute $m^* = (k \cdot m^*) / k$

3.2.1 Why This Works:

- The modified ciphertext $(c_1^*, k \cdot c_2^*)$ is perfectly valid ElGamal encryption
- It decrypts to $k \cdot m^*$ using the standard ElGamal decryption algorithm
- The attacker learns m^* without directly querying the challenge ciphertext
- This violates CCA security completely

3.2.2 Alternative Attack - Re-randomization:

- Pick random s and compute $(c_1^* \cdot g^s, c_2^* \cdot h^s)$
- This encrypts the same message m^* but with different randomness
- Query this "fresh-looking" ciphertext to the oracle
- Directly learn m^*

3.3 How FO Transform Fixes the Malleability Problem

3.3.1 Key Security Properties:

- Randomness Binding: The randomness r is deterministically derived from both the message m and a random value σ using $H_1(m \parallel \sigma)$. This prevents the attacker from controlling the randomness directly.
- Integrity Check: During decryption, we verify that $c_1 = g_1^H(m' \parallel \sigma')$. This ensures the ciphertext hasn't been tampered with, as any modification would likely fail this check.

3.3.2 Attack Prevention in FO-ElGamal:

- **Multiplication Attack:** If attacker creates $(c_1^*, k \cdot c_2^*)$, the decryption gives some $(m'' || \sigma'')$, but the verification $c_1^* \stackrel{?}{=} g_1^H(m'' || \sigma'')$ fails because $k \cdot m^* \neq m^*$ leads to different hash input
- **Re-randomization Attack:** Adding randomness $(c_1^* \cdot g^s, c_2^* \cdot h^s)$ changes c_1^* , but $H_1(m^* || \sigma^*)$ still produces the original randomness, so $c_1^* \cdot g^s \neq g_1^H(m^* || \sigma^*)$ and verification fails
- Any tampering is detected because the consistency check binds the ciphertext components together cryptographically.

3.3.3 Why the Re-encryption Check Works:

The check $c_1 \stackrel{?}{=} g_1^H(m || \sigma)$ ensures that:

- The randomness used to create c_1 matches what would be derived from the decrypted message
- Any modification to either component breaks this relationship
- The hash function makes it computationally infeasible to find collisions that would pass the check

3.3.4 CCA Attack Resistance:

Scenario: Attacker has access to a decryption oracle and wants to decrypt a challenge ciphertext (c_1^*, c_2^*) .

Why FO Transform Prevents This

- If the attacker queries a modified ciphertext (c_1', c_2') to the decryption oracle
- The re-encryption check $c_1' \stackrel{?}{=} g_1^H(m' || \sigma')$ will likely fail for any meaningful modification
- The oracle returns **null**, giving the attacker no useful information
- For the original challenge ciphertext, the attacker can't query it directly (that would be cheating in the CCA game)

4 Why Hash Functions Are Crucial

The hash functions H_1 and H_2 are modeled as random oracles in the security proof:

- H_1 ensures the randomness is unpredictable but verifiable.
- H_2 provides additional randomness extraction.

Together, they bind the message, randomness, and ciphertext components. This binding is what prevents malleability and enables the consistency check that makes CCA attacks fail.